

3) $\binom{N}{2}$ photos 50ms

$$\binom{N}{2} = \frac{N^2 - N}{2} \cdot 50\text{ms} \rightarrow 25(N^2 - N) = \text{CPU Time}$$

If there are $\binom{N}{2}$ photos, each take 50ms,
the total time is $25(N^2 - N)$

$$25(E[N^2]) - E[N] = E[T] \quad \text{Linearity of Expectation}$$

$$\text{Var}(N) = E[N^2] - (E[N])^2 \rightarrow \text{Var}(N) = E[N^2] - (E[N])^2 \quad \text{Variance formula}$$

a) $E(N) = n$ $\text{Var}(N) = 0$ if N is a fixed n , then
 $E(T) = 25(n^2 - n)$ we simply plug n in for N , no variance

b) Binomial (n, p)

$$E[N] = np \quad \text{Var}(N) = np(1-p)$$

$$E[N^2] = np(1-p) + (np)^2 = np - np^2 + n^2 p^2$$

c) Poisson

$$E[N] = \lambda \quad \text{Var}(N) = \lambda$$

$$E[N^2] = \lambda + \lambda^2 = \lambda(\lambda + 1)$$